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What is Data Analytics ?

Data analytics is the systematic process of collecting, cleaning, analyzing, and transforming raw data into meaningful, actionable insights to drive informed decision-making.

Four Main Types of Data Analytics

- **Descriptive Analytics:** Explains what has happened in the past by summarizing historical metrics and trends.
- **Diagnostic Analytics:** Examines why something happened by looking deeper into data correlations and anomalies.
- **Predictive Analytics:** Forecasts what is likely to happen in the future using historical patterns and machine learning.
- **Prescriptive Analytics:** Recommends the best course of action or next steps based on predicted outcomes.

What a Data Analyst Does ?

A data analyst collects, cleans, and interprets raw data to help companies make smart, evidence-based business decisions.

Core Daily Tasks

- **Collect data:** Gather numbers and facts from company databases, surveys, or web tracking.
- **Clean data:** Fix errors, remove duplicates, and organize the messy information so it is safe to use.
- **Analyze data:** Look for trends, patterns, and answers to specific business questions.
- **Visualize data:** Build charts, graphs, and dashboards to make the findings easy to understand.
- **Communicate:** Share reports with managers and team leaders to guide company strategy.

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how analysis supports business decisions ?

Analysis supports business decisions by transforming raw, disconnected data into actionable intelligence, removing reliance on intuition or guesswork. Instead of operating on assumptions, organizations use data analysis to mitigate risks, optimize operations, and discover new revenue streams.



Key Areas Impacted by Data-Driven Choices

1. Strategic Risk Management
2. Hyper-Personalized Customer Experiences
3. Maximizing Operational Efficiency

Identify rows, columns, data types, grain, dimensions, and measures in a dataset ?

To properly analyze a dataset, you need to break it down into its core structural components and define the level of detail it represents.

Rows and Columns

- **Rows (Records):** Horizontal lines in a dataset that represent a single, independent observation, entity, or event.
- **Columns (Fields):** Vertical lines that represent specific attributes, characteristics, or variables associated with each row.

Data Types

Data types define the kind of value a column holds, which dictates how the system stores and processes the information:

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- **Numeric:** Integers (whole numbers) or Decimals/Floats (fractional numbers) used for counts and measurements.
- **Text / String:** Qualitative labels, names, or descriptions.
- **Date / Timestamp:** Temporal values indicating when an event occurred.
- **Boolean:** True/false or yes/no flags.

Grain (Granularity)

- **Definition:** The exact level of detail represented by a single row in the dataset.
- **How to identify:** Complete the sentence: *"Each row represents one [X]."* For example, "one row represents one order line item" or "one row represents one customer per day". Defining the grain first prevents double-counting and aggregation errors.

Dimensions

- **Definition:** Qualitative, descriptive attributes used to filter, group, slice, and dice data.
- **Examples:** Customer names, geographic regions, product categories, or dates. Dimensions answer the *"who, what, where, and when"*.

Measures

- **Definition:** Quantitative, numeric values that can be aggregated (summed, averaged, counted) across dimensions.
- **Examples:** Sales revenue, profit, quantity sold, or transaction amounts. Measures answer the *"how much" or "how many"*